

Tutorial #8

Topics covered:

- Compactness
- Manifolds

Problem 1

- (a) Let $\mathcal{T}$ and $\mathcal{T}'$ be two topologies on the set X ; suppose that $\mathcal{T}' \supset \mathcal{T}$. What does compactness of X under one of these topologies imply about compactness under the other?
- (b) Show that if X is compact Hausdorff under both $\mathcal{T}$ and $\mathcal{T}'$, then either $\mathcal{T}$ and $\mathcal{T}'$ are equal or they are not comparable.

Problem 2

- (a) Show that in the finite complement topology on $\mathbb{R}$, every subspace is compact. (Finite complement topology: $U \subset \mathbb{R}$ is open if and only if: $\mathbb{R} - U$ finite or $U = \mathbb{R}$ or $U = \emptyset$)
- (b) If $\mathbb{R}$ has the topology consisting of all sets A such that $\mathbb{R} - A$ is either countable or all of $\mathbb{R}$, is $[0, 1]$ a compact subspace?

Problem 3

Show that a finite union of compact subspaces of X is compact.

Problem 4

Show that every compact subspace of a metric space is bounded in that metric and is closed. Find a metric space in which not every closed bounded subspace is compact.

Problem 5

Let A and B be disjoint compact subspaces of the Hausdorff space X . Show that there exist disjoint open sets U and V containing A and B , respectively.

Problem 6

Show that if $f : X \rightarrow Y$ is continuous, where X is compact and Y is Hausdorff, then f is a closed map (that is, f carries closed sets to closed sets).

Problem 7

Show that if Y is compact, then the projection $\pi_1 : X \times Y \rightarrow X$ is a closed map.

Problem 8

Let X be a metric space with metric d ; let $A \subset X$ be nonempty.

- (a) Show that $d(x, A) = 0$ if and only if $x \in \bar{A}$.
- (b) Show that if A is compact, $d(x, A) = d(x, a)$ for some $a \in A$.
- (c) Define the ϵ -neighborhood of A in X to be the set

$$U(A, \epsilon) = \{x \mid d(x, A) < \epsilon\}$$

Show that $U(A, \epsilon)$ equals the union of the open balls $B_d(a, \epsilon)$ for $a \in A$.

- (d) Assume that A is compact; let U be an open set containing A . Show that some ϵ -neighborhood of A is contained in U .
- (e) Show the result in (d) need not hold if A is closed but not compact.

Problem 9

Show that a connected metric space having more than one point is uncountable.

Problem 10

Let X be limit point compact.

- (a) If $f : X \rightarrow Y$ is continuous, does it follow that $f(X)$ is limit point compact?
- (b) If A is a closed subset of X , does it follow that A is limit point compact?
- (c) If X is a subspace of the Hausdorff space Z , does it follow that X is closed in Z ?

We comment that it is not in general true that the product of two limit point compact spaces is limit point compact, even if the Hausdorff condition is assumed. But the examples are fairly sophisticated.

Problem 11

Let (X, d) be a metric space. If $f : X \rightarrow X$ satisfies the condition

$$d(f(x), f(y)) = d(x, y)$$

for all $x, y \in X$, then f is called an isometry of X . Show that if f is an isometry and X is compact, then f is bijective and hence a homeomorphism. **Hint:** If $a \notin f(X)$, choose ϵ so that the ϵ -neighborhood of a is disjoint from $f(X)$. Set $x_1 = a$, and $x_{n+1} = f(x_n)$ in general. Show that $d(x_n, x_m) \geq \epsilon$ for $n \neq m$.

Problem 12

On $\mathbb{R}^{n+1} \setminus \{0\}$ define the equivalence relation

$$x \sim y \Leftrightarrow \text{there is a nonzero real constant } \lambda \text{ such that } x = \lambda y.$$

The real projective n -space is defined by $\mathbb{RP}^n = (\mathbb{R}^{n+1} \setminus \{0\}) / \sim$. Show that $\mathbb{RP}^n$ is a manifold.

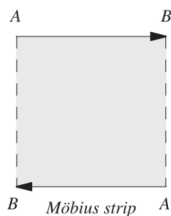
Hint: build charts (U_i, φ_i) with

$$U_i = \{[x_1, \dots, x_{n+1}] \in \mathbb{RP}^n \mid x_i \neq 0\},$$

$$1 \leq i \leq n+1.$$

Problem 13

The Möbius strip can be constructed as the quotient of $(0, 1) \times [0, 1]$ by the equivalence relation determined by $(x, 0) \sim (1 - x, 1)$, see the following figure.



Prove that the Möbius strip is a manifold, by showing that the following two maps provide an atlas:

$$\begin{aligned} \varphi_1 :]0, 1[\times]0, 1[&\rightarrow]0, 1[\times]0, 1[, \quad \varphi_1(x, y) = (x, y) \\ \varphi_2 : (]0, 1[\times ([0, \epsilon[\cup]1 - \epsilon, 1])) / \sim &\rightarrow]0, 1[\times] - \epsilon, \epsilon[, \\ \varphi_2([x, y]) &= \begin{cases} (x, y) & \text{if } y \in [0, \epsilon[\\ (1 - x, y - 1) & \text{if } y \in]1 - \epsilon, 1] \end{cases} \end{aligned}$$